a

True for each sample

True for the distribution

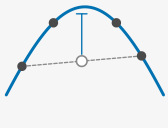
C1 Local linear



$A\mathbf{x} = 0$

$\nabla\cdot\mathbf{u} = 0$, $\nabla\cdot\mathbf{B} = 0$
Homogeneous linear boundary conditions

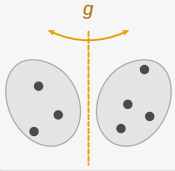
C2 Local nonlinear



$c(\mathbf{x}) = 0$

Stationary PDE residual

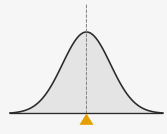
C3 Symmetry



$g_{\#}\mu = \mu$

$\mathbf{B} \mapsto -\mathbf{B}$
Rotation R_π (VKS)

C4 Distributional statistic



$\mathbb{E}_\mu[\mathcal{L}\varphi] = 0$

Energy budget
$\mathbb{E}[\varepsilon] = \mathbb{E}[P]$

Matched mechanism

Potentials $\mathbf{B} = \nabla\times\mathbf{A}$ *or* Leray projection	Nonlinear projection *or* Penalty, guidance	Equivariant v_θ *and* Invariant base distribution	Stationarity loss *or* Exponential tilting

b

Energy budget (C4) *Illustrative shapes*

